# Week 46, Tuesday Session

## Flipped classroom: your questions on trees and forests, then Project 3 – read the text, choose a path, and start drafting your proposal

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# How today works

## The format (as every Tuesday)

Mondays are lectures; Tuesdays belong to your questions and to concrete work in groups. Today the concrete work is the start of **Project 3**, the last and freest project of the course.

1. **First ∼20 min:** your open questions from Monday's lecture – the impurity measures, the CART split, the Gini and entropy arithmetic, our own tree code, Eq. (7.16), bagging, the out-of-bag error, the random forest – discussed in plenum.
2. **∼15 min:** the text of Project 3, walked through: the deliverables, the updated rules on LLMs, the three paths, what is common to all of them.
3. **∼15 min, in your groups:** read the three paths and decide which one interests you most – and why.
4. **∼35 min, in your groups:** start drafting a **proposal**: problem, data, methods, plan. Other data sets than the ones we suggest are welcome. We circulate.
5. **Last 5 min:** wrap-up. **The aim for today is a sketch of 3–5 slides** of what you want to do.

## The sketch is the weekly exercise of week 47 – the last one

The exercise text (`exercisesweek47.ipynb`) is published already. You present the proposal in the final flipped classroom on Tuesday November 17 and hand in the finished version by **Friday November 20**. Details on the last frame.

# Picking up from Monday

## Monday in one sentence each

**A tree partitions the feature space into axis-aligned boxes and predicts a constant in each, grown top-down and greedily. A split is scored by the weighted impurity of its children, Eq. (7.9) – MSE for regression, Gini index or entropy for classification – and CART keeps the largest gain, Eq. (7.10), over every feature and every midpoint; the ten points and the ride data showed the arithmetic, sixty lines of Python reproduced `scikit-learn`. A full-depth tree has zero training error and large variance. $\mathrm{var}(\bar{f}) = \rho\sigma^2 + (1-\rho)\sigma^2/B$ (7.16): bagging shrinks the second term, the random forest lowers $\rho$ with $m \approx \sqrt{p}$ features per split (7.18), and the out-of-bag error (7.17) comes for free.**

## Questions we expect – ask them

- Why do the Gini index and the entropy almost always pick the same split, and when could they differ?
- Why only the midpoints between consecutive values? What does a tree predict outside the range of the training data?
- Why does adding trees to a forest never overfit, when adding depth to one tree does?
- Why $m \approx \sqrt{p}$ – and what happens at $m = 1$ and $m = p$?
- Is the out-of-bag error a substitute for cross-validation?
- On the cancer data logistic regression won. So when are forests the right choice?

## Everything is in `week46.ipynb`

Impurities, the CART scan, `build_id3`, the `DecisionTree`, `BaggingClassifier` and `RandomForestClassifier` classes, every number on the slides.

# Short answers, in case they are needed

- **Gini vs entropy.** Both are strictly concave in the class proportions and nearly proportional for two classes ($S/2$ tracks $G$); they rank splits the same way unless two candidates are very close. They differ in *values*, never in the tree, on both of Monday's examples.
- **Midpoints.** The criterion is constant between consecutive sorted values, so there is nothing to gain in between; outside the training range a tree returns the value of the outermost leaf **– no extrapolation**, and a forest inherits this.
- **Error rate as criterion.** Piecewise linear: it does not reward a split that creates a pure child unless the majority vote changes ($(400, 400)$ example). Concave measures do, and a pure node never needs splitting again.
- **More trees never overfit.** Eq. (7.16): $B$ only enters the term $(1 - \rho)\sigma^2/B$, which decreases; the bias is untouched. Depth increases the variance $\sigma^2$ of each tree instead.
- **$m$.** $m = p$ is bagging ($\rho$ high); $m = 1$ picks the split feature at random ($\rho$ low, $\sigma^2$ high). $\sqrt{p}$ is a rule of thumb; the cancer scan showed $\rho$ from 0.68 to 0.81 and a flat out-of-bag accuracy – cross-validate $m$ when it matters.
- **OOB vs CV.** For bagged ensembles, yes, as an estimate of the test error of *that* ensemble; it is not a replacement for a held-out test set once you have tuned on it.
- **When forests.** Tabular data with many mixed features, interactions, no good parametric form, little time for feature engineering; not for one smooth feature, not for extrapolation, not when a linear model is already at the Bayes rate.

# Project 3: the frame

## Deliverables and deadline

As in Projects 1 and 2: a report of about 5000 words and 10–15 figures, written as a scientific report; the GitHub repository with `Code`, `README` and the PDF; delivered as a group in Canvas. **Deadline: December 11 (midnight).**

## The updated rules on LLMs – read them

The guidelines were changed this week. For the **text of the report** only *contribution level 1* is allowed: spell-checking, grammar, minor phrasing. Having an LLM draft, rewrite or restructure a section is **not acceptable use**. For the **code**, levels 0–4 still apply, with the inline `LLM-assisted` tag. The declaration after the conclusions is required (−5 points if missing) and now states the level (0 or 1) for every text section and the level for every code file.

## What is new: you choose

Projects 1 and 2 prescribed data and methods. Project 3 offers **three paths**, and you pick one – or propose your own problem, after writing to us.

## What is the same for every path

Own code or libraries (say which); the quality measures of Projects 1–2 where they apply; hyperparameters on training/validation data and the test set once; seeds and documented settings; a link to the literature; a critical assessment. The report structure is the one you know.

## Where the text is

`doc/Projects/2026/Project3/` in the course repository: PDF, LaTeX and a notebook with the same text and two starter cells.

# Path A: Solving differential equations with neural networks

## The problem

The one-dimensional diffusion equation $u_{xx} = u_t$ on $[0, 1]$ with $u(x, 0) = \sin(\pi x)$ and $u(0, t) = u(1, t) = 0$ – or another ordinary or partial differential equation agreed with us (wave, Poisson, coupled ODEs, the matrix-eigenvalue ODE).

## The parts

a) the scheme, the analytical solution, the stability condition;

b) the explicit scheme at $\Delta x = 1/10$, $1/100$, two times, errors;

c) neural networks: a trial function with **hard-coded** conditions, and a **PINN** with soft conditions; against the scheme and the analytical solution;

d) nodes, layers, activations, collocation points, loss weights, optimizer;

e) critical assessment.

## Who this is for

You liked weeks 42–43 (Chapter 9), you know finite differences from FYS3150/4150 or MAT-MEK4270, and you want a problem with an exact answer to compare against.

## What you need

Your Project-2 network, or Keras/PyTorch/JAX; automatic differentiation for $\partial_x$, $\partial_{xx}$ and $\partial_t$ of the network *output with respect to its inputs*; patience with training (PINNs are fussy about the learning rate and the loss weights).

## Typical risk

Spending all the time on getting one PINN to converge and none on part d). Get the hard-coded version working first.

# Path B: A classification study with several methods

## The problem

One multiclass (or rich binary) classification data set that is *not* MNIST, analyzed with **at least three** methods of the course, among them **at least one neural network** (FFNN or CNN) and **at least one tree-based ensemble** (random forest, gradient boosting, XGBoost); logistic regression as the cheap baseline.

## Default data set: Fashion-MNIST

70 000 grey-scale $28 \times 28$ images of clothing in ten classes, the MNIST format exactly – your Project-2 code runs unchanged – but harder: roughly 0.84 (logistic) to 0.91–0.94 (CNNs). Shirt, T-shirt, pullover and coat get confused, so the confusion matrix means something.

## Other data

CIFAR-10; UCI tabular sets (covertype, adult, wine quality, spambase); sequence data for an RNN; Kaggle; your own field – write to us first.

## The parts

a) the data in context; b) the theory of your methods; c) implementation, hyperparameter search with cross-validation, times and parameter counts; d) results on one test set – accuracy with standard error, confusion matrix, per-class precision/recall/$F_1$, one-vs-rest ROC and AUC, misclassified samples, importances or filters – against the literature; e) critical assessment.

## Who this is for

You want the workflow of an applied study: data, baseline, several methods, honest comparison, published numbers to beat or to explain.

## Typical risk

Three methods run once each with default settings and a table of three accuracies. The points are in the tuning, the uncertainties and the analysis of *why* the ranking is what it is.

# Path C: The statistics and mathematics behind the methods

## The idea

Synthetic data where the *truth is known* – a function plus noise of known $\sigma^2$, a Gaussian mixture with known Bayes error $\Phi(-\Delta/2)$ – so that bias, variance, the Bayes rate, the correlation between trees and the bias of an error estimator can all be **derived and then measured**.

## The parts

a) the derivations (bias-variance, OLS/Ridge expectation and covariance, Bayes error, logistic regression as maximum likelihood with the Fisher information, Eq. (7.16), what each resampling estimator estimates); b) linear models and networks: measured against closed-form bias and variance, the Ridge path, width/epochs/$L_2$, optional double descent; c) classifiers against the Bayes error, calibration, class imbalance; d) ensembles against Eq. (7.16), $\rho$ against $m$, bias-variance of tree/bagging/forest; e) CV, LOO, bootstrap, .632 and OOB *as estimators*: their bias and variance, the one-SE rule, nested CV; f) critical assessment.

## Who this is for

You enjoyed the derivations more than the data sets, and you want to see Chapters 2, 3, 5, 7 and 8 confirmed – or refined – by experiments you control completely. Mostly your own code (projects 1 and 2) plus `scikit-learn`.

## What it is not

Not a literature project and not a big-data project: the data are cheap to generate, the computing is in the repetitions (many training sets, many bootstrap samples).

## Typical risk

Measuring everything and deriving nothing – or the reverse. The report must pair each measurement with the formula it tests, and say where they disagree and why.

# How we read a Project 3 report

## The same grading as before

Abstract, introduction, theory and methods, implementation and tests, results and discussion, conclusions and perspectives, the LLM declaration, references. Figures with captions that make a point; tables with uncertainties; code that we can rerun from the README.

## What the freedom adds

- A clear **question** on page one: what did you set out to find?
- A **justified choice** of data and methods – why these, what is the baseline, what would a sceptic want to see.
- A **comparison** with something external: analytical solution, published benchmark, the formula, the Bayes rate.
- A **critical assessment** that goes beyond "method X was best".

## The seven take-aways in the text

Method fits problem; a number needs a baseline and an error bar; model selection is not model assessment; flexibility costs variance; averaging removes variance, not bias; verification before results; reproducibility. Use them as the checklist for your conclusions.

## Scope

About five weeks, next to other courses. A narrow question answered properly beats a wide one half done. The proposal you start today is where you set the scope – and where we can tell you if it is too much or too little.

# Group work, step 1 (∼15 min): which path?

## Read the three paths in the project text, then answer in your group

1. Which path do you find most **interesting** – and is that the same as the one you find easiest? Say both.
2. What do you **already have**? Project-1 regression and bootstrap/CV code, the Project-2 network and its MNIST pipeline, today's tree and forest code, finite-difference code from other courses, Keras or PyTorch experience.
3. What would you have to **learn** for each path (PINNs and autodiff of network inputs; gradient boosting and CNN tuning; Fisher information and the error-estimator study)? Is that learning the kind you want?
4. **Compute and data access**: do you have a laptop that can train a CNN on 60 000 images, or a GPU in Colab? Is the data set you have in mind downloadable without registration and allowed for this use?
5. Who in the group would do what?

## Decide, and write the decision down in one sentence

"We choose path __ because __; the main thing we must learn is __." If you cannot decide between two paths, note what would settle it and ask us when we come by.

# Group work, step 2 (∼35 min): draft the proposal – the sketch

## The aim for today: 3–5 slides, roughly one per item

1. **The question and the data.** One sentence stating what you want to find out. The data: what, where from, how many samples and features, what the target is, who has analyzed them before (one or two references). For path A: which equation, which conditions, what the analytical or reference solution is. For path C: which synthetic setup, with which known quantities.
2. **Methods and baseline.** Which two or three methods, which is the baseline, what is own code and what is a library, which quality measures, how the hyperparameters will be chosen.
3. **Plan and risks.** A timeline from now to December 11 with the division of labour; the one thing most likely to go wrong and what you do if it does; the computing you need.
4. **(Optional) The figures you expect.** Sketch the two or three key figures of the report before you have the data: what is on the axes, what the curves should look like if you are right.

## Format

Slides (Beamer, PowerPoint, Keynote, Google Slides – anything that becomes a PDF). Sketchy is fine today; the finished version is next week's exercise.

# If you consider a data set of your own: the checklist

## Practical

- Can it be downloaded today, without registration, in a format you can read (csv, npz, images in folders)? Is the licence compatible with coursework?
- Size: enough samples to train, validate and test (thousands rather than hundreds); small enough to iterate on a laptop.
- Labels: clean, documented classes or targets; a known class balance.
- A documented or canonical train/test split, or a plan for making one.

## Scientific

- At least one published analysis to compare with – a benchmark table, an article, a Kaggle leaderboard with documented methods.
- A question that the methods of the course can answer: tabular $\rightarrow$ trees and ensembles against logistic regression and a network; images $\rightarrow$ CNN against the rest; sequences $\rightarrow$ RNN; a known equation $\rightarrow$ path A.
- Something you understand: domain knowledge makes the discussion.

## Then

Tell us when we come by, or write to us before the week-47 session. A data set that passes this checklist is as welcome as Fashion-MNIST.

# What a sketch can look like (an example for path B)

## Page 1 – question and data

***Can a gradient-boosting model on raw pixels compete with a small convolutional network on Fashion-MNIST, and at what cost?***
Fashion-MNIST: 70 000 $28 \times 28$ grey-scale images, 10 balanced classes, canonical 60 000/10 000 split; Xiao et al. 2017, benchmark table on the Zalando repository (logistic $\approx 0.84$, GB $\approx 0.88$, CNN 0.91–0.94).

## Page 2 – methods and baseline

Baseline: logistic regression (Project 2, zero hidden layers). Methods: (1) our Project-2 FFNN, 784–128–10, Adam, $L_2$; (2) `HistGradientBoostingClassifier` / XGBoost; (3) Keras CNN of week 44. Hyperparameters by 5-fold CV on 10 000 training images, final models on all 60 000, test set once. Measures: accuracy $\pm$ SE, confusion matrix, per-class $F_1$, one-vs-rest ROC/AUC, training time, parameter count.

## Page 3 – plan and risks

Week 47: data pipeline, baseline, FFNN rerun (A). Week 48: boosting tuning (B), CNN tuning (A). Week 49: results, figures, misclassified samples (all). Week 50: writing, LLM declaration, README, freeze on Dec 9. Risk: CNN tuning too slow on laptops $\rightarrow$ Colab GPU, or a fixed small architecture with only the learning rate and dropout tuned.

## Page 4 – expected figures

Accuracy of all four models against training-set size (log x-axis); confusion matrices side by side with the shirt/T-shirt/pullover/coat block highlighted; AUC per class; accuracy against training time.

## Why this is a good sketch

A question, numbers to beat, a baseline, a plan with names and dates, a risk with a fallback, and figures imagined before the data. Yours can be rougher today.

# Prompts for the other two paths while you draft

## Path A

- Diffusion equation as given, or another equation? If another: do you have an analytical or a trusted numerical reference solution?
- Which library for the networks, and how will you get $u_{xx}$ and $u_t$ of the network – `jax.grad` twice, `torch.autograd`, `tf.GradientTape`?
- Which architectures and activations go into the part-d) scan, and how many collocation points?
- What is your measure of error – against the analytical solution at two times, over the domain, as a function of training time?

## Path C

- Which regression function (Runge 1D, 2D, Franke) and which $\sigma$? Which Gaussian mixture, and will the Bayes boundary be linear or curved?
- How many training sets and bootstrap samples can you afford per experiment? (Hundreds are needed; plan the loops.)
- Which derivations of part a) will you write out in full, and which will you cite?
- Which optional items (double descent, boosting) will you include, if any?

## For everyone

Which figure of the report is the one you would show to prove your point? Draw it now, with axes and expected curves.

# Wrap-up: the sketch becomes the weekly exercise of week 47

## The last weekly exercise

Finish the sketch you started today into a **proposal of 3–5 slides**: question and data, methods and baseline, plan and risks, (optional) the expected figures. If you propose a data set of your own, include the checklist items (source, size, licence, split, literature). Add the short LLM declaration as in the earlier weekly exercises. The exercise text is already in the week-47 folder; hand in the final version as a PDF in Canvas, one per group, by **Friday November 20 (midnight)** – after you have presented it and heard our reaction.

## The final flipped classroom, Tuesday November 17

Each group presents its proposal – the 3–5 slides as they stand that day, 3–5 minutes – we and the other groups react, and you leave with a plan that we have seen. Revise it with that feedback before Friday. This is the last chance to redirect before the project deadline; use it.

## Until then

- Read the full text of your path, including the summary of requirements and the take-aways.
- Download the data and look at them before you promise anything.
- Paths A and B: get the baseline running (explicit scheme; logistic regression). Path C: the data generators and the bias-variance loop.

## Monday November 16: boosting

AdaBoost, gradient boosting, XGBoost, and boosting any loss with automatic differentiation (Sections 7.11–7.15) – the method that attacks the bias that today's ensembles leave alone, and a natural third method for path B.